

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report: Use the Ipconfig Command

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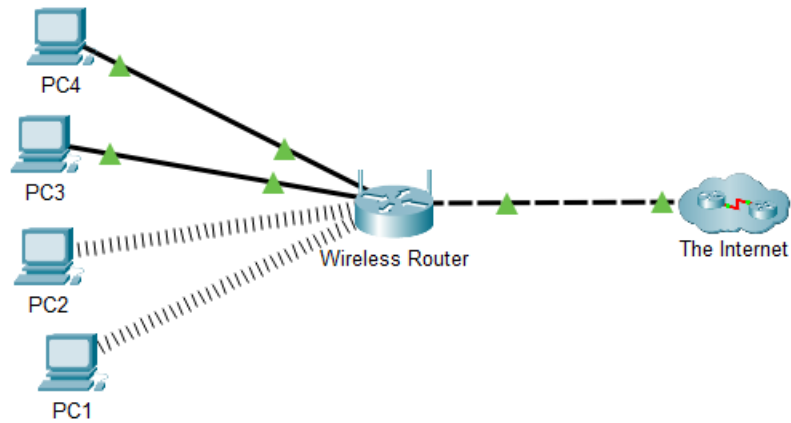
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1. Objectives

- Use the ipconfig command to identify an incorrectly configured PC.

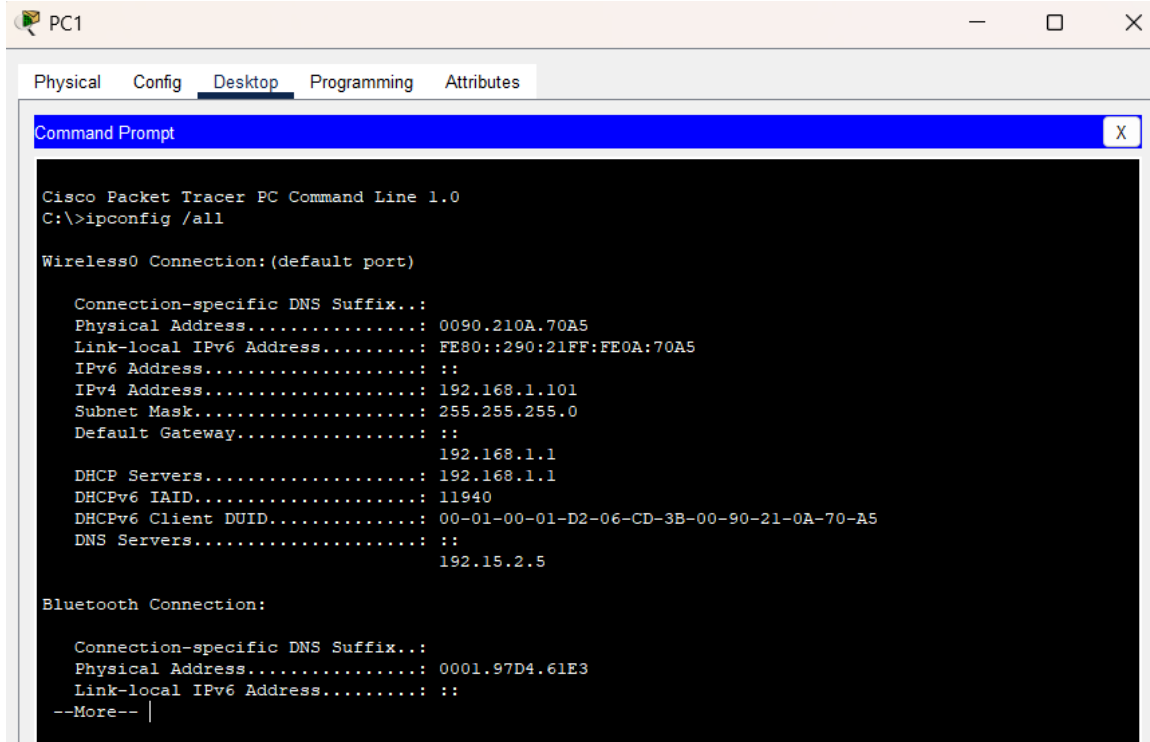
2. Background / Scenario

A small business has four PCs configured with static IP addresses in the 192.168.1.0/24 network. One PC cannot reach the internet or access the website www.cisco.pka. You must determine which PC is misconfigured using the ipconfig /all command.



3. Part 1: Verify Configurations

- Open Command Prompt on each PC.
- Run the command: `ipconfig /all`
- Record the IP address, subnet mask, and default gateway for all PCs.



The screenshot shows a PC window titled 'PC1' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the output of the `ipconfig /all` command, detailing network configuration for both Wireless0 and Bluetooth connections.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

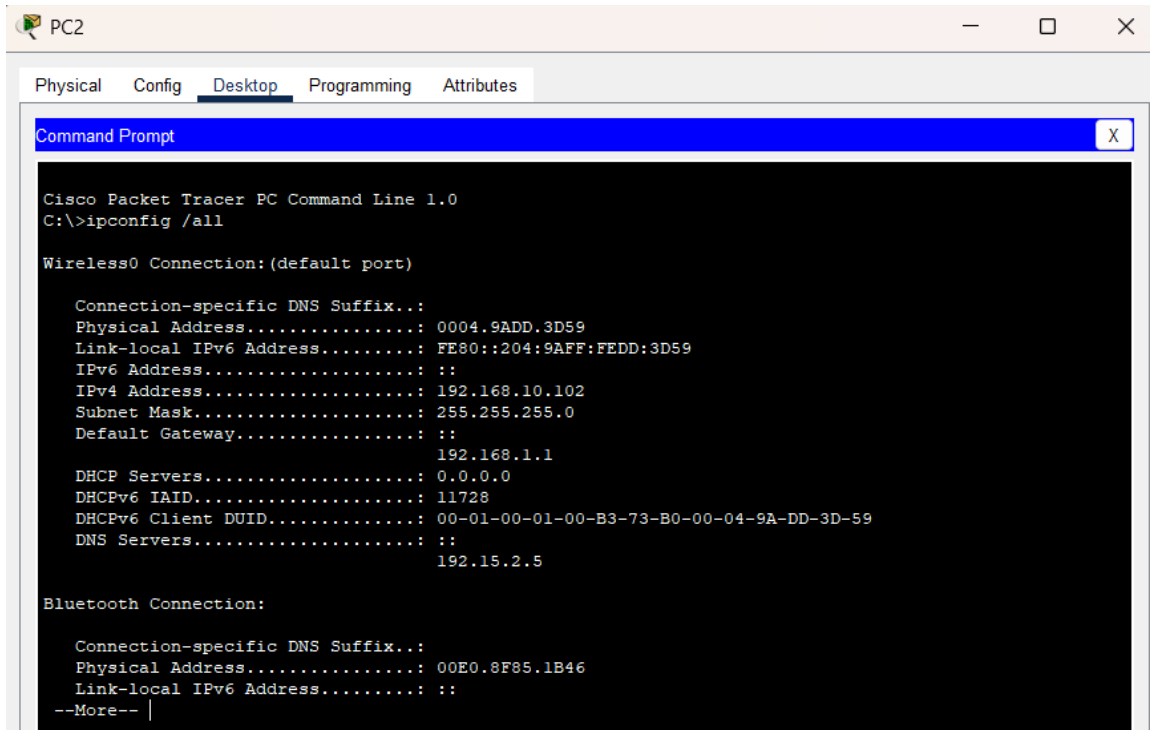
Wireless0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address.....: 0090.210A.70A5
Link-local IPv6 Address.....: FE80::290:21FF:FE0A:70A5
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.101
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.1.1
DHCP Servers.....: 192.168.1.1
DHCPv6 IAID.....: 11940
DHCPv6 Client DUID.....: 00-01-00-01-D2-06-CD-3B-00-90-21-0A-70-A5
DNS Servers.....: ::
                        192.15.2.5

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 0001.97D4.61E3
Link-local IPv6 Address.....: ::
--More-- |
```

PC1 ipconfig



PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

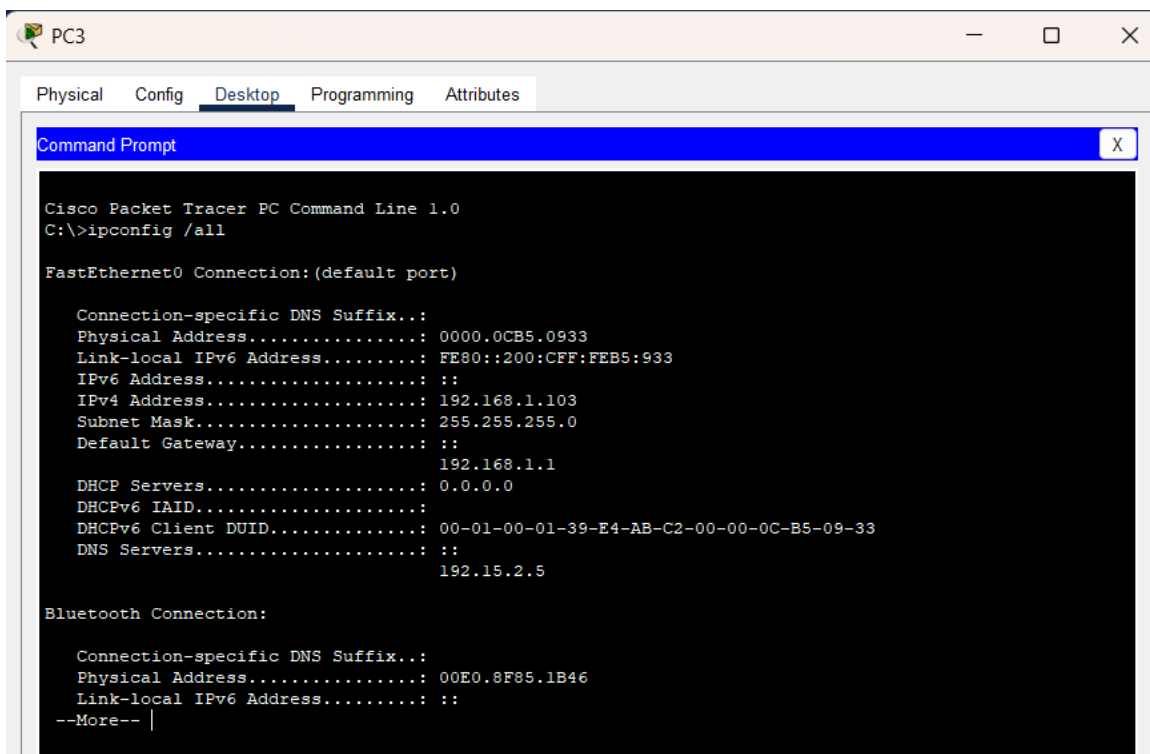
Wireless0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address.....: 0004.9ADD.3D59
Link-local IPv6 Address.....: FE80::204:9AFF:FEDD:3D59
IPv6 Address.....: ::
IPv4 Address.....: 192.168.10.102
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.1.1
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....: 11728
DHCPv6 Client DUID.....: 00-01-00-01-00-B3-73-B0-00-04-9A-DD-3D-59
DNS Servers.....: ::
                        192.15.2.5

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 00E0.8F85.1B46
Link-local IPv6 Address.....: ::
--More-- |
```

PC2 ipconfig



PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address.....: 0000.0CB5.0933
Link-local IPv6 Address.....: FE80::200:CFF:FEB5:933
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.103
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.1.1
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-39-E4-AB-C2-00-00-0C-B5-09-33
DNS Servers.....: ::
                        192.15.2.5

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 00E0.8F85.1B46
Link-local IPv6 Address.....: ::
--More-- |
```

PC3 ipconfig

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0050.0FC6.0C67
    Link-local IPv6 Address.....: FE80::250:FFF:FEC6:C67
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.1.104
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        192.168.1.1
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-4C-67-1C-24-00-50-0F-C6-0C-67
    DNS Servers.....: ::
                        192.15.2.5

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address.....: 0001.C743.D6E4
    Link-local IPv6 Address.....: ::
    --More--
  
```

PC4 ipconfig

Analysis of IP Information

Correct configuration should follow the pattern:

- IP Address: 192.168.1.x
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

Any deviation identifies the misconfigured PC.

Below is the completed table summarizing the IP configuration observed on each PC during the lab.

PC	IP Address	Subnet Mask	Default Gateway	Status
PC1	192.168.1.101	255.255.255.0	192.168.1.1	Correct
PC2	192.168.10.102 (incorrect network)	255.255.255.0	192.168.1.1	Incorrect
PC3	192.168.1.103	255.255.255.0	192.168.1.1	Correct
PC4	192.168.1.104	255.255.255.0	192.168.1.1	Correct

Identified Misconfigured PC

After comparing all four PCs:

PC2 is the misconfigured device.

It used:

- **Wrong network:** 192.168.10.x instead of 192.168.1.x

PC2 contained incorrect IP settings and was the reason for the connectivity issue. After correcting the IP address to match the 192.168.1.0/24 network, the PC regained proper connectivity. This demonstrates how essential the **ipconfig /all** command is for diagnosing Layer 3 addressing problems.

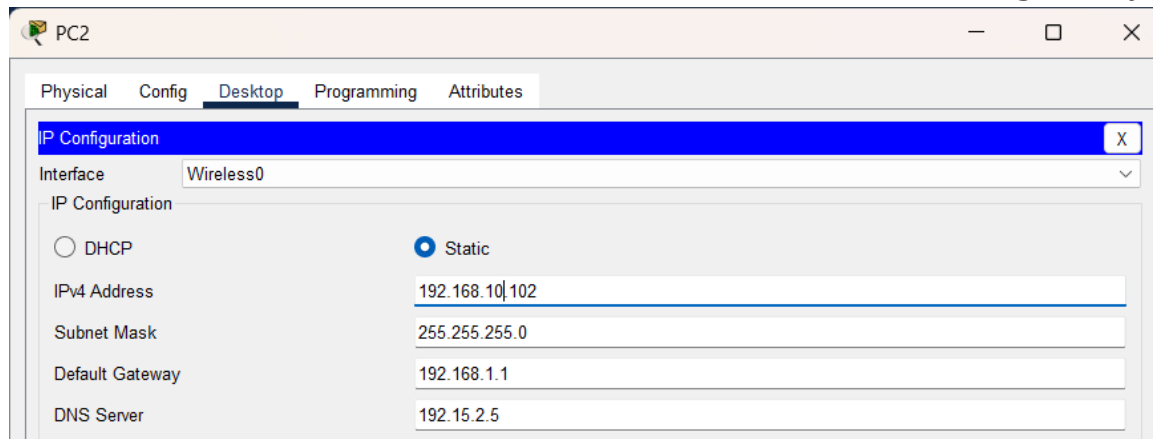
4. Part 2: Correct Any Misconfigurations

a. Select the PC that has incorrect IP information.

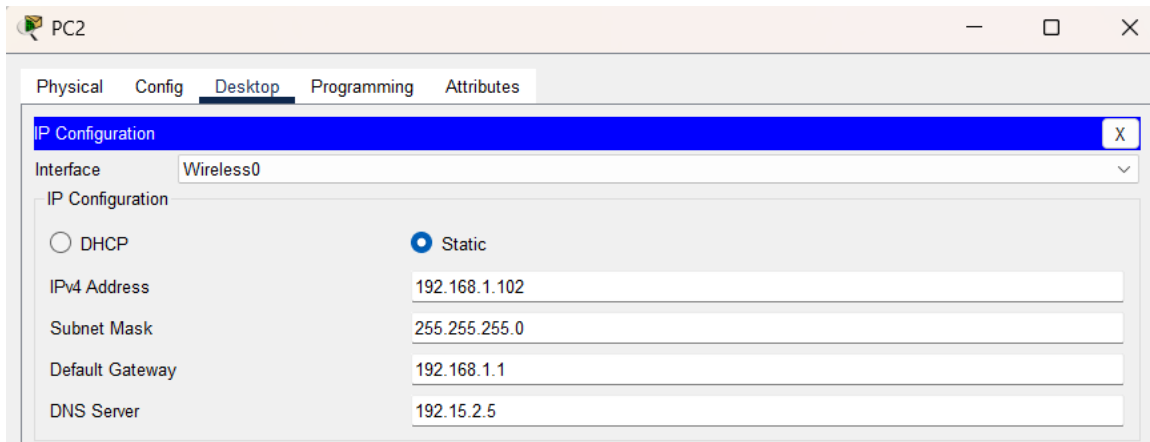


b. Open Desktop > IP Configuration.

c. Enter correct values for IP address, subnet mask, and default gateway.



Before correct ip settings



Corrected IP Settings

5. Observations & Conclusion

After correcting the addressing issue, the PC should successfully reach www.cisco.pka. The `ipconfig` command is a primary troubleshooting tool for identifying IP configuration issues.